

Kuan-Chieh Hsu

✉ khsu@bnl.gov | 🏠 <https://www.bnl.gov/staff/khsu> | 🌐 <https://kuan-chiehhsu.github.io/>

Research Interest

High-performance computing; Heterogeneous/parallel computing; computer architecture/system; Scientific/engineering computing; AI for system

Education

University of California, Riverside

PhD in Computer Science and Engineering

Riverside, CA

2019 - 2024

- GPA: 3.78 / 4.0
- Thesis: General-Purpose Tensor Processors: A Broader Computational Paradigm

National Cheng Kung University

Tainan, Taiwan

M.S. in Computer and Communication Engineering

2014 - 2016

- GPA: 4.0 / 4.0
- Thesis: Performance Prediction Model on HSA-Compatible General-Purpose GPU System

National Cheng Kung University

Tainan, Taiwan

B.S. in Electrical Engineering

2010 - 2014

- GPA: 3.91 / 4.0

Experiences

Research Associate

Upton, NY

Computing and Data Sciences, Brookhaven National Laboratory

2024 - current

- performance modeling, scientific computing workflow

Research Assistant

Riverside, CA

Extreme Scale Computer Architecture Laboratory, UCR

2019 - 2024

- **SHMT: Simultaneous and Heterogeneous Multithreading**
 - Proposed a new parallel execution and programming model to allow simultaneous execution of heterogeneous accelerators including Jetson Nano's GPU and Google's Edge TPU to achieve 1.95x speedup and 51% energy saving compared to optimal GPU baseline for a set of parallelism-rich applications in C++.
 - the open source code: <https://github.com/escalab/SHMT>
- **GPTPU: Accelerating Applications using Edge Tensor Processing Units**
 - Developed a programming framework and the system library for Google's EdgeTPU AI/ML accelerators to achieve 2.46x speedup and 40% energy saving compared to high-performance CPU cores for a set of matrix applications using C++.
 - Redesigned the algorithm and implemented the library function for general matrix multiplications using 2D convolutions to more efficiently use the EdgeTPU hardware and achieve 41.2x speedup using C++.
 - the open source code: <https://github.com/escalab/GPTPU>

Graduate Research Program Intern

Oak Ridge, TN

Oak Ridge National Laboratory (ORNL)

Fall and Winter 2023

- Explored the general-purpose computing using tensor processing units for future grid simulation.

Performance Team Intern

Sunnyvale, CA

Cerebras System

Summer 2022

- Developed MLIR-based infrastructure including IR readers to support compilation flow integration with analytical performance models.

System Architect Intern

San Jose, CA

Samsung Strategy and Innovation Center

Summer 2019

- Developed a set of deep learning kernel functions on an FPGA-based hardware acceleration platform using OpenCL via SDAccel environment.

Research Assistant (transferred out after the one year PhD program studying)

Raleigh, NC

Extreme Scale Computer Architecture Laboratory, NCSU

2018 - 2019

- **Autonomous Near-Data Processing on Intelligent SSD**
 - Implemented Financial and linear algebra applications in Python with C-wrapper Python functions to invoke GPU kernels. The implementation offers our proposed Python runtime to achieve the same level of performance of C runtime.

Research Assistant

Taipei, Taiwan

Data Insights Research Lab, Institute of Information Science, Academia Sinica

2017

- Proposed a typhoon trajectory forecasting framework that augments limited real-world data and produces Euclidean distance prediction error within 100 kms using TensorFlow-based model.
- Organized the conference as a core staff including session scheduling and hardware facilities arrangement: 4th Taiwan Artificial Intelligence/Data Science Conference in Academia Sinica, Taiwan (2000+ participants)

Research Assistant

Tainan, Taiwan

Computer Architecture and System Lab, NCKU

2014 - 2016

- Implemented an early-staged heterogeneous system simulation including scalable memory subsystem with hierarchical cache system, NoC module, and memory controllers for OpenCL and OpenGL applications using hardware-style abstraction by C/C++.
- Designed modularized class materials for a Computer Organization course based on a full-system CPU simulator with programs implementing Fibonacci sequence to orient students with function calls using MIPS.

Summer Intern

Hsinchu, Taiwan

Nuvoton Technology corp.

Summer 2014

- Participated in early-staged design of an universal protocol IC project via summarizing timing, port definition and frequency discrepancy among involved protocols including I2C, SMBus and UART.
- Implemented data processing automation and visualization in R language to avoid human intervention on weekly generated measurement data.

Publications

- [1] Tasnuva Chowdhury, Tadashi Maeno, Fatih Furkan Akman, Joseph Boudreau, Sankha Dutta, Shengyu Feng, Adolffy Hoisie, **Kuan-Chieh Hsu**, Raees Khan, Jaehyung Kim, Ozgur O. Kilic, Scott Klasky, Alexei Klimentov, Tatiana Korchuganova, Verena Ingrid Martinez Outschoorn, Paul Nilsson, David K. Park, Norbert Podhorszki, Yihui Ren, John Rembrandt Steele, Frederic Suter, Sairam Sri Vatsavai, Torre Wenaus, Wei Yang, Yiming Yang, and Shinjae Yoo, “Machine Learning-Driven Predictive Resource Management in Complex Science Workflows, *International Journal of Modern Physic*, 2025 [\[DOI\]](#)
- [2] Sankha Dutta, Ozgur O. Kilic, Tania Korchuganova, Paul Nilsson, Sairam Sri Vatsavai, **Kuan-Chieh Hsu**, David Park, Joseph Boudreau, Tasnuva Chowdhury, Shengyu Feng, Raees Ahmad Khan, Jaehyung Kim, Scott Klasky, Tadashi Maeno, Verena Ingrid Martinez Outschoorn, Norbert Podhorszki, Yihui (Ray) Ren, Fred Suter, Wei Yang, Yiming Yang, Shinjae Yoo, Alexei Klimentov and Adolffy Hoisie, “Error Analysis of Globally Distributed Workflow Management System” the 7th Annual Workshop on Emerging Parallel and Distributed Runtime Systems and Middleware (IPDRM), 2025 [\[DOI\]](#)
- [3] Sairam Sri Vatsavai, Raees Ahmad Khan, **Kuan-Chieh Hsu**, Ozgur O. Kilic, Paul Nilsson, Tania Korchuganova, David Park, Sankha Dutta, Yihui (Ray) Ren, Joseph Boudreau, Tasnuva Chowdhury, Shengyu Feng, Jaehyung Kim, Scott Klasky, Tadashi Maeno, Verena Ingrid Martinez Outschoorn, Norbert Podhorszki, Fred Suter, Wei Yang, Yiming Yang, Shinjae Yoo, Alexei Klimentov and Adolffy Hoisie, “CGSim: A Simulation Framework for Large Scale Distributed Computing Environment” the 16th International Workshop on Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems (PMBS), 2025 [\[DOI\]](#)
- [4] **Kuan-Chieh Hsu**, Sairam Sri Vatsavai, Ozgur O. Kilic, Tania Korchuganova, Paul Nilsson, Sankha Dutta, Yihui (Ray) Ren, David Park, Joseph Boudreau, Tasnuva Chowdhury, Shengyu Feng, Raees Ahmad Khan, Jaehyung Kim, Scott Klasky, Tadashi Maeno, Verena Ingrid Martinez Outschoorn, Norbert Podhorszki, Fred Suter, Wei Yang, Yiming Yang, Shinjae Yoo, Alexei Klimentov and Adolffy Hoisie, “Data Management System Analysis for Distributed Computing Workloads,” the 11th International Workshop on Data Analysis and Reduction for Big Scientific Data (DRBSD), 2025 [\[DOI\]](#)
- [5] Phani R V Marthi, Soumyajit Gangopadhyay, **Kuan-Chieh Hsu**, Jongchan Choi, Nilanjan Ray Chaudhuri, and Suman Debnath, “RE-INTEGRATE EMT Simulation Software: DAE Solvers and Automation,” *IEEE PES General Meeting 2025, Austin, TX* [\[Paper\]](#)
- [6] **Kuan-Chieh Hsu**, Jongchan Choi, and Suman Debnath, “Sparse Linear Solvers for Large-scale Electromagnetic Transient Simulations,” in *50th Annual Conference of the IEEE Industrial Electronics Society, (IECON) 24*’ [\[Paper\]](#)
- [7] **Kuan-Chieh Hsu**, and Hung-Wei Tseng, “Simultaneous and Heterogenous Multithreading: Exploiting Simultaneous and Heterogeneous Parallelism in Accelerator-Rich Architectures,” in *IEEE Micro*, vol. 44, no. 4, pp. 11-19, 24’ [\[Paper\]](#)
- [8] **Kuan-Chieh Hsu**, and Hung-Wei Tseng, “Simultaneous and Heterogenous Multithreading,” in *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 23’ [\[Paper\]](#)
- [9] Yu-Chia Liu, **Kuan-Chieh Hsu**, and Hung-Wei Tseng, “Rethinking Programming Frameworks for In-Storage Processing,” *Proceedings of the 60th ACM/IEEE Design Automation Conference (DAC) 23*’ [\[Paper\]](#)
- [10] **Kuan-Chieh Hsu**, and Hung-Wei Tseng, “GPTPU: Accelerating Applications using Edge Tensor Processing Units,” *In International Conference for High Performance Computing, Networking, Storage and Analysis, SC 21*’ [\[Paper\]](#)
- [11] Yun-Chi Huang, **Kuan-Chieh Hsu**, Wan-shan Hsieh, Chen-Chieh Wang, Chia-Han Lu, and Chung-Ho Chen, “Dynamic SIMD Re-convergence with Paired-Path Comparison,” *the IEEE International Symposium on Circuit and System (ISCAS)*, 16’ [\[Paper\]](#)
- [12] Heng-Yi Chen, Chung-Ho Chen, Yun-Chi Huang, **Kuan-Chieh Hsu**, and Chen-Chieh Wang, “An HSAIL ISA conformed GPU platform,” *Applied System Innovation - Proceedings of the International Conference on Applied System Innovation, (ICASI) 15*’ [\[Paper\]](#)

Technical Skills

Professional	C/C++, CUDA, Python, OpenCL, MATLAB, OpenMP, and R
Version control	git
Systems	Linux, Mac, Windows
Deep Learning	Keras, TensorFlow, Pytorch
Educational	Scratch